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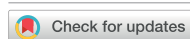
To cite this article: Johanna Mehnert *et al* 2026 *Environ. Res.: Energy* **3** 025002

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OPEN ACCESS

RECEIVED

18 December 2025

REVISED

20 March 2026

ACCEPTED FOR PUBLICATION

26 March 2026

PUBLISHED

7 April 2026

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Long-term scenarios and energy system impacts of technological carbon dioxide removal deployment in Finland

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Keywords: carbon dioxide removal, energy system, Finland, scenarios, BECCS, DACCS, biochar

Supplementary material for this article is available [online](#)

Abstract

This study analyzed energy system impacts of technological carbon dioxide removal (CDR) deployment in Finland. We modeled long-term scenarios up to 2050 for four CDR technologies: bioenergy with carbon capture and storage (BECCS), biochar soil amendment, direct air carbon capture and storage (DACCS), and enhanced weathering of mining rock waste (EW). An integrated energy economic model compiled using the TIMES-model generator was used to produce cost-minimal development scenarios for Finland's energy system, including CDR technologies. Three scenarios were modeled: one without a specific CDR target and two with low- and high CDR targets. The results show that CDR targets primarily affect the deployment of biochar, BECCS, and DACCS, whereas EW potential was fully utilized in all scenarios. Most of the growth in the CDR deployment originated from DACCS, while BECCS and biochar remained relatively insensitive to the CDR targets. The limited availability of sustainable biomass and its allocation across competing sectors played a central role in shaping the complex interactions between BECCS, biochar, and DACCS in the Finnish energy system. Regarding national net greenhouse gas (GHG) emissions, the findings indicate that the net-zero GHG target is more easily achieved under any binding CDR target than without one. However, substantial CDR deployment would still be required for Finland to achieve climate neutrality even by 2050. The impacts of CDR technologies extend more broadly throughout the energy system than shown in earlier studies due to interconnections between the energy-consuming and producing sectors.

1. Introduction

The EU's climate policy aims for net-zero greenhouse gas (GHG) emissions by 2050 [1]. To reach the net-zero and the intermediate targets of a 55% reduction by 2030 and a 90% reduction³ by 2040 requires contributions from all EU member states.

Finland is a large but sparsely populated EU member state in Northern Europe, whose energy system is characterized by the use of renewables and nuclear power. Finland's GHG emissions without land use, land use change and forestry (LULUCF) were 38.8 Mt CO_{2eq} in 2024, representing a decrease of 46% since 1990 [2]. Most of the decrease in GHG emissions is due to emission reductions in the energy sector, where coal has been phased out already in 2025, four years ahead of the target schedule [3, 4]. The use of peat and hard coal has decreased over time, while wood-based biomass accounted for 31%, nuclear 24%, and other renewables 19% of primary energy consumption in early 2025 [2, 5]. Despite this, the energy sector (NFR 1), including transport, remains the largest source of GHGs in Finland. Within

³ In November 2025, after the modeling work had been conducted, the European Parliament voted to back the Council's proposal for an 85% domestic emission reduction target by 2040, instead of the 90% target that was earlier suggested.

the energy sector, domestic transport is responsible for 37% of the emissions in 2024, having the slowest pace of emission reductions since 1990 [2].

To help speed up the transition to a climate-neutral society, Finland updated its climate law in 2022 by setting targets for climate neutrality in 2035 and climate negativity afterwards [6]. Additionally, the climate law sets GHG emission reduction targets for EU emission trading system (EU-ETS) and effort-sharing sectors to bring down gross GHG emissions and to support the transition to fossil-free energy sources. However, meeting these targets will increasingly depend on how Finland balances its remaining emissions with carbon sinks. Historically, Finland had a significant net carbon sink in the land use sector (LULUCF), over $-20 \text{ Mt CO}_{2\text{eq}}$ until 2016 [7]. However, in recent years, this carbon sink has been lost due to increases in timber harvesting, a slowdown of tree growth, and increasing soil emissions [7]. The LULUCF sector has turned into a source of emissions of about $13.5 \text{ Mt CO}_{2\text{eq}} \text{ a}^{-1}$ [7]. In the most recent reference scenario modeled for Finland's energy and climate strategy and medium-term climate policy plan, Finland does not achieve climate neutrality by 2035 largely due to the LULUCF sector emissions [8]; the gap to the target depends strongly on assumptions about timber harvesting, which have been assumed to increase significantly due to the suspension of wood import from Russia.

This growing uncertainty in the land sector highlights the importance of technological carbon dioxide removal (CDR) as a complementary mean to ensure Finland's progress toward climate neutrality and climate negativity. Although the Finnish climate law aims for net negative GHG emissions in the long term, Finnish regulations do not specify any numerical targets for sectoral emission reductions nor CDR.

CDR deployment in Finland has been mostly studied by focusing on BECCS potential (bioenergy with carbon capture and storage) [9–12]. As biomass plays a major role in Finland's energy system, the theoretical potential for BECCS is high: currently biogenic CO_2 emissions from large industrial sites amount to $28 \text{ Mt CO}_2 \text{ a}^{-1}$ [10]. Pulp and paper mills and large district heating power plants are ideal point sources for biogenic CO_2 capture [10, 12]. The demand for domestic wood-based biomass has traditionally been driven by the pulp and paper industry, which has historically used approximately half of the harvested timber [13].

The recent cessation of wood imports from Russia has resulted in increasing demand and market prices for domestically sourced wood [13, 14]. In recent baseline scenarios, timber harvesting and energy use of wood-based biomass are projected to increase slightly in the long term [8, 15]. Both reports acknowledge the potential for BECCS, but recognize the lack of national and EU-level incentives and too low carbon prices in the EU-ETS as a barrier for BECCS deployment [8]. Challenges in BECCS deployment include decentralized heat production, low-capacity point sources, low flue-gas CO_2 concentrations, and process-efficiency losses when capturing biogenic CO_2 [10, 12]. Biogas production could provide high concentration of CO_2 in flue gas, but volumes may be too low for cost-efficient capture [10].

Beyond BECCS, other biomass-based CDR options such as biochar are also relevant in the Finnish context due to the country's extensive forestry and agricultural sectors. Residual biomass streams from forestry, agriculture, and biowaste provide an opportunity for pyrolysis and biochar production [16, 17]. Biochar soil amendment offers a method for creating negative emissions by storing the carbon in the soil over a long period [16]. Applying biochar in agricultural soils has been shown to have multiple co-benefits in addition to carbon sequestration [16]. Biochar application in managed forests could also be a viable option, as it has been shown to increase both belowground and aboveground carbon sinks in forests [18]. Currently, biochar is produced in Finland for several applications with estimated annual production volume of 10 kt of biochar [19]. However, soil amendment is not yet among the use applications due to the lack of profitability [19, 20]. Survey results suggest that increasing carbon prices could improve profitability [20, 21]. No regional study has yet included biochar soil amendment as a CDR option in long-term scenarios, although biochar production for other uses has been modeled [8, 15]. We identify that biochar-related CDR potential is shaped by competition with non-CDR applications, and the availability and cost of suitable feedstock.

Another technologically distinct CDR method for high-latitudes is direct air capture of CO_2 and storage direct air carbon capture and storage (DACCS). DACCS represents an energy-intensive alternative to BECCS and is projected to have a larger role in Europe's decarbonization as it approaches the net-zero targets in 2050 [22–24]. DACCS costs arise from the high energy demand needed to produce heat for dissolving or releasing captured CO_2 from solvents or sorbents [23, 25, 26]. In the context of decarbonization, DACCS heat demand can be met by several low-carbon energy sources, including renewable electricity-based heating, nuclear heat, waste heat, or other renewable heat options [23]. In Finland, relatively low electricity prices could decrease the operating costs of direct air capture (DAC) [11, 23]. In addition, studies show that a cold environment could help reduce the energy consumption

of solid sorbent DAC [27, 28]. However, estimated energy consumption and costs of different DAC technologies vary widely, which brings another layer of uncertainty to modeling future DAC deployment [11, 22, 23, 25, 26, 29, 30].

Enhanced weathering (EW) offers a further CDR option with potential links to Finland's mining industry. The most recognized EW method involves spreading crushed silicate-containing rocks on soils [29]. In the Nordic context, EW remains under study, although a recognized potential exists in cement waste in addition to mining waste rocks [8]. A study modeling CDR deployment on the European level included EW in the CDR portfolio, but the results show little or no EW deployment in Finland [24].

Captured CO₂ needs to be stored permanently to realize net negative emissions. Finland has no known geological CO₂ storage [31]. The nearest feasible geological CO₂ storage lies in the North Sea near the Norwegian coastline [10, 11, 31]. This implies long transport distances involving ship and land transport in the absence of pipelines. The European CO₂ storage network and storage capacity are both under development and may become bottlenecks for large-scale CDR deployment [10, 31].

Carbon capture and utilization (CCU) is an alternative to geological storage, although it cannot deliver negative emissions as the captured CO₂ will be released back to the atmosphere via various routes. The climate benefits from CCU could arise through the substitution of fossil products. Captured CO₂ can be used to make products such as electro fuels (e-fuels, also called renewable fuels of non-biological origin, RFNBO) [32]. The absence of domestic geological CO₂ storage sites could make CCU economically attractive for Finland. However, CCU requires large investments in CO₂ processing, expanded renewable electricity, and hydrogen production [9]. Increasing EU-wide demand for RFNBO due to the Renewable Energy Directive (REIII), ReFuelEU Aviation, and ReFuelEU Maritime [33–35] is expected to increase the demand for biogenic and atmospheric CO₂, especially since fossil CO₂ from EU-ETS installations cannot be used for RFNBOs after 2040 [36].

Several technologically feasible CDR and CCU options exist in the Finnish context, as presented. Yet, their potential roles and interactions with the energy and land systems remain poorly understood. Current legislation, particularly EU-level mandates for RFNBOs, is expected to drive CCU deployment, but no dedicated legislative framework exists yet to promote CDR deployment. Because CDR methods differ significantly in their energy requirements, feedstock needs, and coupling with heat, electricity, and land systems, understanding how large-scale CDR deployment would influence Finland's future energy system is essential for informed policy and investment planning.

Until today, to the authors' knowledge, there has been no peer-reviewed analysis of how a diverse set of CDR technologies could be deployed cost-efficiently in Finland, and how such deployment would interact with the national energy system. A study on negative emission technologies and practices in the EU included Finland; however, the results were analyzed only on the EU-level, giving no further insights into national CDR deployment potential [24]. Technical reports and white papers regarding BECCS deployment [10] or its economic impacts [9] have been published, but the analyses neglected other potential technological CDR methods such as DACCS, biochar, and EW. Another technical report [11] introduces practical views on CDR and CCU markets, impacts on GHG accounting and policy development in Finland, but lacks long-term scenarios and cross-sectoral impacts of CDR deployment. Until recently, the integrated energy system assessments [15, 37, 38] produced for Finland's national climate and energy strategies did not include a comprehensive assessment of the potential for CDR technologies in Finland. However, the latest long-term scenarios [8, 30] include BECCS and DACCS, but do not consider biochar and EW as CDR methods.

This study aims to fill the gap by modeling long-term CDR deployment in Finland with several feasible CDR technologies including BECCS, DACCS, biochar, and EW. We will use an integrated energy economic model to develop cost-optimized scenarios for energy system development with varying CDR targets, together with regulatory CCU targets. In our analysis, we focus on the impacts of CDR deployment on the energy system and identify cross-sectoral impacts, synergies and potential barriers. Some of the findings of the Finnish case study are also applicable in other regions. This is reflected in the discussions and conclusions section.

2. Methodology

Long-term scenarios for technological CDR deployment in Finland are modeled using an integrated energy economic model for the period 2025–2050. We consider BECCS, DACCS, biochar, and EW as CDR technologies. Additionally, we model CCU technologies that are interconnected to CDR deployment via technological CO₂ capture. We analyze the deployment of CDR and related energy system impacts widely, focusing on electricity generation, heat production, biomass use, and e-fuel use.

Table 1. Assumed volumes of timber harvesting, and categorization of available feedstock in Finland.

Million solid cubic meters (Mm ³)	2010	2020	2030	2040	2050	References
Material wood total	52.0	58.5	70.4	70.1	70.1	Historical years
Wood products from saw timber	14.2	16.3	21.8	21.8	21.8	2010–2020: [13, 41, 42]
Sawdust from saw timber	7.7	8.8	11.7	11.7	11.7	Future years 2030–2050: [15]
Pulpwood	30.1	33.5	36.9	36.6	36.6	
Energy wood total	10.3	12.9	17.1	17.3	17.3	
Stemwood	7.7	10.3	11.5	11.4	11.2	
Logging residues	2.6	2.6	5.6	5.9	6.1	

2.1. The model

The model is an integrated energy economic model compiled with the TIMES-model generator [24, 39]; it is based on the public JRC-EU-TIMES model [40]. The model solves a single cost-minimal solution for the modeled time horizon and regions, which represents a set of technologies that respect the given constraints and fulfill the energy demands across sectors. The model includes CO₂ emissions from all energy-producing and consuming sectors of society (NFR 1–2), but excludes GHG emissions from agriculture, land use, and waste management (NFR 3–5). In the following sections, we describe the key assumptions behind our scenarios that drive, affect, and limit CDR and CCU deployment in the model. Detailed cost assumptions for BECCS, DACCS, and biochar technologies are introduced in the supplementary information (SI), and the entire model database is available at [39].

2.2. Biomass feedstock

BECCS and biochar deployment are directly constrained by the availability of suitable biomass feedstock. The modeled biomass feedstocks include various energy crops, agricultural residues, willow and poplar, municipal biowaste, sewage sludge, and several forestry-based biomass feedstocks. We have partially updated the biomass feedstock availability forecast for Finland in the model. The effects of environmental conditions, such as climate change impacts on crop yields, are not included in the exogenous biomass availability assumptions.

The amount of available forestry feedstock is based on the baseline scenario PEIKKO-with existing measures (WEM)-P in [15]. Projection of timber harvesting in the future is derived from the demand for wood for various applications (table 1); in the case of Finland, the pulp and paper industry is a major wood consumer. The scenario has an underlying assumption that timber harvesting is not regulated, and the demand for pulp and paper increases in the future [8, 15]. Timber harvesting is categorized as industrial roundwood and as energy wood harvests. Annual industrial roundwood harvesting is expected to increase from 67.5 Mm³ (million solid cubic meters of wood) to 70.1 Mm³ by 2050, while energy wood harvesting will somewhat increase from 10.3 Mm³ to 11.2 Mm³ [15]. That equals a total harvesting volume of around 78–82 Mm³/a for the modeled future years. Energy wood is further categorized as stemwood and logging residues, while industrial roundwood includes pulpwood and saw timber, of which quantities are reported in [15]. The model includes categories pulpwood, stemwood for energy, logging residues for energy, and sawdust for energy.

The quantity of sawdust generated in the saw mills from saw timber is estimated based on statistics: we assume 35% of saw timber becomes sawdust, while 65% ends up in wood products [13]. Material wood is partially converted into energy wood in the model via a chemical pulp production process, in which black liquor is the main output.

We use a conversion factor of 7.2 PJ Mm⁻³ to convert cubic meters of wood into useful energy. The conversion factor corresponds to a lower heating value of 7.2 MJ kg⁻¹ of semi-moist wood weighing 1000 kg m⁻³ as a solid cubic meter of wood.

Agricultural feedstock potentials originate mostly from the JRC-EU-TIMES model assumptions [40]. The amount of available agricultural residues for energy use is assumed to stay at 18 PJ a⁻¹ [40], while biogas production from anaerobic digestion of manure increases up to 6.5 PJ a⁻¹ by 2050 [15]. Energy crop potentials are assumed to be zero for all except willow, which increases to 7.4 PJ a⁻¹ by 2050 [40]. Lastly, biowaste and sewage sludge potentials originate from the JRC-EU-TIMES-model assumptions: biowaste feedstock increases from 9.4 PJ a⁻¹ in 2020 to 17 PJ a⁻¹ by 2050, while sewage sludge increases up to 4 PJ a⁻¹ by 2050 [40].

In addition to domestic biomass feedstock, trade of biomass feedstock is allowed within the modeled EU31 regions: in 2025, a maximum of 15 PJ of imports to Finland is allowed, decreasing to 10 PJ by 2035. To allow flexibility on domestic biofuel production, we allow trade of biofuels within the EU market (up to 60 PJ a⁻¹ by 2030, increasing to 100 PJ a⁻¹ by 2050) and some extra-EU imports of biofuels (up to 45 PJ a⁻¹).

2.3. CDR technologies

The modeled CDR technologies include BECCS, DACCS, biochar soil amendment, and EW.

In the model, black liquor is produced as a pulp and paper industry side product from material wood feedstock: black liquor can be used for BECCUS (bioenergy with carbon capture, utilisation and storage) in two applications: retrofitted boilers for industrial heat, and biofuel production. Agricultural residues, forestry residues, primary stemwood, and willow can be used in biofuel production with or without BECCUS.

Two DAC technologies are modeled. Liquid solvent DAC (L-DAC) is modeled with a mix of heat sources, with the option of using electricity only as the heat source. S-DAC is modeled with a mix of heat sources. DAC processes are not constrained by any other variable directly, but the CO₂ transport costs, geological CO₂ storage costs and capacity, as well as availability of fossil-free energy affect the amount of DACCS indirectly. If CO₂ storage capacity limits DACCS in a scenario run, the model still has an option to utilize the captured CO₂ in e-fuel production. Detailed cost assumptions for BECCS and DACCS technologies are introduced in the SI.

The modeled biochar production process is fully updated: costs and energy balances were adjusted using calculated costs, and energy flows from example pyrolysis plants in the Finnish context presented in [43, 44]. In addition, currently established production capacities and expert estimates of realistic scale-up in Finland were used to define the upper limits for biochar production. Present biochar production in Finland is estimated at 10 kt a⁻¹, and a new biochar production plant will increase this capacity by 60 kt a⁻¹ [15, 19]. Total biochar production is expected to grow approximately tenfold by 2030 [19]. In the longer term, we assume production scaling up to a maximum of 1 Mt a⁻¹ [19].

Only residual biomass streams, agricultural residues, forestry residues, and municipal biowaste are allowed as feedstock for biochar production used for soil amendment and CDR. Notably, current biochar production in Finland is largely directed to other applications, such as replacing coal in industry [8]. Because our model does not include non-CDR use cases for biochar, nor trade of biochar between regions, we do not force the existing production capacity to be utilized. Instead, the model invests in new biochar production capacity for soil amendment when doing so is cost-efficient.

EW of mining waste rocks is modeled as another long-term material storage of CO₂. EW as technology is still in the development phase, and information regarding CO₂ sequestration potential and related costs is scarce [30]. Hence, we use expert estimates as our upper limits of CO₂ sequestration defined at 0.5 Mt CO₂ by 2030, increasing to 1 Mt by 2035, and 2 Mt by 2050 [45]. The cost of EW is assumed to be a constant 140 EUR (2020)/tCO₂ [24].

2.4. CO₂ storage and utilization

The nearest geological CO₂ storage is located in the North Sea; Finland does not have geological storage capacity [10, 11, 31]. Hence, we assume that captured carbon needs to be imported by ships to the Norwegian coast of the North sea for long-term storage. Estimated CO₂ transport costs vary greatly depending on the geographical location within Finland: the country's position at the Baltic Sea makes CO₂ transport from the coast cheaper than from inland [10]. There is no infrastructure ready for CO₂ transport from inland [10], and we do not assume nor model investments into CO₂ pipelines from inland: all CO₂ transport is done via existing road and sea routes. However, we assume that most of the CO₂ capture is done at or near the coastline due to significant cost efficiency differences when compared to inland CO₂ capture and transport [10, 46]. Also, we assume that the exogenously defined cost of CO₂ transport is brought down by increasing transport volumes in the 2040s and 2050s. Using the average values of the ranges provided in [45], we set the CO₂ transport and storage cost for Finland to 88 EUR(2020)/tCO₂ from 2025, and let it decrease to 73 EUR(2020)/tCO₂ by 2030, 57 EUR(2020)/tCO₂ by 2040, and 41 EUR(2020)/tCO₂ by 2050.

Geological CO₂ storage in Europe is updated to match the amounts of planned annual CO₂ capacities in [31]. By 2030, the total planned capacity within EU31 is approximately 161 Mt CO₂ a⁻¹, increasing to 191 Mt CO₂ a⁻¹ by 2035. After this, we assume a rapid linear increase in storage capacity reaching 775 Mt CO₂ a⁻¹ by 2040 and 2.6 Gt CO₂ a⁻¹ by 2050 [47]. National CO₂ injection capacities per storage type are scaled up from 2035 so that they stay below the total regional cumulative CO₂ storage capacities [47], resulting in a total modeled cumulative CO₂ storage capacity until 2050 of 20.4 Gt CO₂,

Table 2. Description of the modeled scenarios.

Name	Description	CDR + CCU target	CCU target
S0	Reference scenario, no forced CDR	No direct target	Minimum RFNBO production volumes and the required CO ₂ capture amounts originate from RED III, ReFuelEU Aviation and Maritime [33–35].
S1	Scenario with forced CDR	5 Mt CO ₂ in 2035 10 Mt CO ₂ in 2040 15 Mt CO ₂ in 2050	
S2	Scenario with forced CDR	5 Mt CO ₂ in 2035 15 Mt CO ₂ in 2040 25 Mt CO ₂ in 2050	

Table 3. Targets for renewable fuels in transport fuels in the scenarios.

Year	Fuel type	Aviation fuel	Road transport diesel	Road transport gasoline	Road transport CNG	Maritime heavy fuel oil
2030	Renewables total	6%–9%	39%–41%	7%–30%	≤100%	6%–7%
	RFNBOs	1%–1.5%	1%–1.5%	1%–1.5%	≤0.5%	1%–2%
2035	Renewables total	20%–23%	30%–79%	8%–33%	≤100%	12%–100%
	RFNBOs	5%–6%	1%–5%	2%–5%	≤5%	2%–7%
2040	Renewables total	34%–37%	16%–100%	10%–40%	≤100%	18%–100%
	RFNBOs	10%–11%	2%–5%	4%–12%	≤12%	3%–11%
2045	Renewables total	42%–45%	11%–100%	12%–63%	≤100%	24%–100%
	RFNBOs	15%–16%	7%–11%	6%–35%	≤35%	4%–16%
2050	Renewables total	70%–73%	11%–100%	14%–78%	≤100%	26%–100%
	RFNBOs	35%–36%	9%–13%	8%–50%	≤50%	5%–36%

which remains significantly lower than the estimate of 117 Gt CO₂ in [47]. Norway's CO₂ storage capacity affects Finland's opportunity to deploy CCS if the demand for geological storage in the EU is high. Norway's CO₂ storage capacity is limited to 54, 64, 101, and 201 Mt CO₂ a⁻¹ by 2030, 2035, 2040, and 2050, respectively [31, 47].

Instead of geological storage, captured CO₂ can be used in the production of e-fuels. The model includes several power-2-X applications, which can convert captured CO₂ to e-kerosene, e-diesel, e-gasoline, or e-methane. The model separates the use cases for captured fossil and non-fossil CO₂. Fossil CO₂ capture and utilization is allowed until 2040 [36], after which fossil CO₂ can only be stored in geological storage in the model.

2.5. Scenarios

We model three scenarios: one without a specific CDR target and two scenarios with higher and lower CDR targets (table 2). The main goal is to explore the impacts of various levels of CDR deployment in Finland on the energy system under the EU climate and energy policy. We base our CDR deployment levels on the previous national scenario results regarding CO₂ capture in Finland [8, 30]. In the recent with additional measures-scenario [8], the total CO₂ capture started from 3 Mt CO₂ a⁻¹ in 2035, increasing up to 16 Mt CO₂ a⁻¹ by 2050. In [30], the four long-term scenarios, representing alternative futures, had largely varying results, ranging from 0.5–5.5 Mt CO₂ a⁻¹ in 2035 up to 2.5–20 Mt CO₂ a⁻¹ in 2050. Based on these ranges, we defined feasible targets for CDR deployment ranging from 15 Mt CO₂ up to 25 Mt CO₂ by 2050.

Scenario S0 represents a reference scenario in which there is no target for CDR. Only CCU-targets apply in S0, as the modeled regulations remain constant in our scenarios (table 3). CDR may be deployed in S0 nevertheless, if it becomes cost-efficient in relation to GHG mitigation in the model. Also, CCU targets create demand for domestic CC.

In scenarios S1 and S2, CDR targets are set indirectly due to model structure: the model is given a minimum target for the sum of CDR + CCU. This means the sum of CO₂ captured from bioenergy and DAC, and CO₂ sequestration by biochar soil amendment and EW. The model needs to fulfill the

same minimum CCU requirements as in S0. After fulfilling the minimum CCU target, it has the freedom to choose between more CCU, BECCS, DACCS, biochar, or EW for a cost-optimal solution of each scenario.

CCU targets have been set as minimum RFNBO-targets for transport fuels according to three EU Regulations: RED III [33], ReFuelEU Aviation [34], and ReFuelEU Maritime [35]. The targets for different transport modes are presented in table 3.

2.6. Policy assumptions

GHG mitigation is driven by regional and EU-level climate and energy policy. Finland's climate law requires a reduction of gross GHG emissions of 60% by 2030, 80% by 2040, and urges to reach 95% reduction by 2050 compared to 1990 [6]. Finland's climate neutrality target in 2035 is not modeled because the scope of the model does not include emissions from agriculture (NFR 3), LULUCF (NFR 4), and waste management (NFR 5).

In addition to the regional climate law [6], Finland faces the EU climate policy [1], which is assumed to stay in its current format throughout the modeled time horizon. Renewable energy requirements set by the RED III [33] are considered as minimum requirements to be fulfilled by all regions. Minimum levels of RFNBO in aviation, maritime, and road transport fuels are set as in RED III, ReFuelEU Aviation, and ReFuelEU Maritime until 2035 [33–35], after which we allow rapid increases in the shares of RFNBO fuels reaching a maximum of 100% by 2050. EU ETS1 and ETS2 [48, 49] face increasing carbon prices rising from 80 EUR (2020)/tCO₂ in 2030 up to 161 EUR (2020)/tCO₂ by 2050 based on the WEM prices by the EC [50]. Decarbonization of the vehicle fleet is driven by CO₂ limits for new cars, vans, buses, and trucks sold in the EU member states [51, 52]. In the model, we assume that the EU collectively reaches net zero CO₂ emissions from energy-producing and consuming activities by 2050; however, international aviation and navigation are excluded from the net-zero target given to the model [1]. As the use of fossil energy sources decreases, driven by the net-zero emission target, we assume that the share of non-CO₂ GHGs from combustion decreases accordingly, and most of the energy-related GHGs are CO₂ in the long term.

2.7. Sensitivity analyses

We conduct a series of sensitivity analyses to detect the sensitivity of the main cost assumptions related to CDR deployment. The chosen cost variables are: DAC investment costs, EW carbon sequestration cost, and biomass feedstock costs. The costs are varied $\pm 25\%$ in their own sensitivity analysis scenarios labeled as 'high' indicating +25% change, and 'low' indicating –25% change. The sensitivity analyses are conducted for the main scenario S0, which represents the cost-efficient baseline without forced CDR.

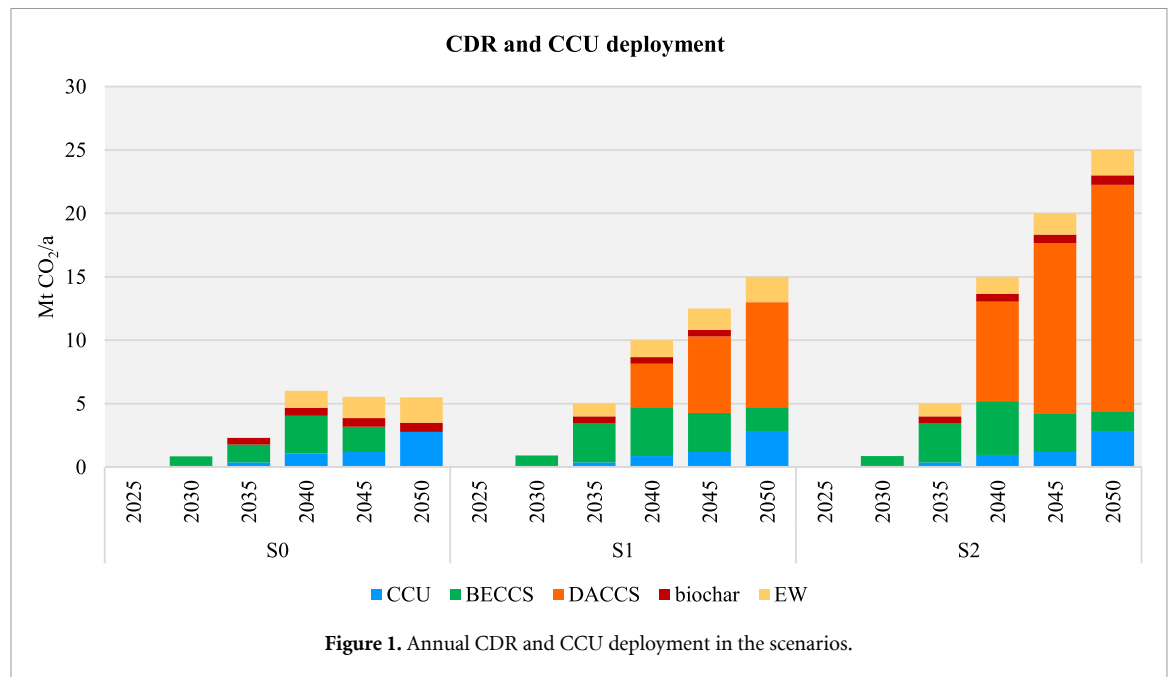
3. Results

3.1. CDR deployment

CDR is deployed in all scenarios alongside CCU, with variable portfolio and deployment trends (figure 1). Every modeled CDR technology is deployed cost-efficiently at least in one of the scenarios. CDR deployment was driven by CDR targets in S1/S2 (table 2), while CCU deployment was driven by the minimum RFNBO requirements for road transport, maritime, and aviation fuels in the EU (table 3).

In S0, CDR deployment begins in 2030 at 0.7 Mt CO₂ a^{–1}, reaches its peak of 4.9 Mt CO₂ a^{–1} in 2040, and then declines to 2.7 Mt CO₂ a^{–1} by 2050. Of this, BECCS accounts for 73% in the 2030s, over 46% in the 2040s, and its share drops to 0% in 2050, when all the CO₂ captured by BECCS is used for CCU. All CO₂ capture in the S0 scenario is from biogenic CO₂; DAC technology is not deployed in S0 because its use is relatively more expensive than other modeled CDR technologies. The decline in the use of BECCS means that the share of biochar and EW from total CDR deployment grows significantly in the long term in the cost-optimal scenario. The full biochar deployment potential is not used in any of the scenarios, which indicates that biochar deployment has been constrained by feedstock availability and/or process-related costs. However, EW is deployed up to the allowed upper limit, which indicates that EW deployment is cost-efficient with the current cost assumptions.

When examining the results from alternative scenarios S1 and S2, where the total CDR + CCU amount is forced to a certain level, clear changes are observed in the deployment portfolio and trends when compared to the S0 scenario. The total CDR + CCU amounts follow the minimum constraints set for the scenarios (S1: 10 Mt CO₂ 2040, 15 Mt CO₂ 2050; S2: 15 Mt CO₂ 2040, 25 Mt CO₂ 2050); deploying more than the required minimum was not cost-efficient from the entire system's perspective.



EW is utilized almost to the full extent within the allowed upper limit similarly as in S0, after which the required CDR + CCU is achieved through biochar, BECCS, and DACCS deployment.

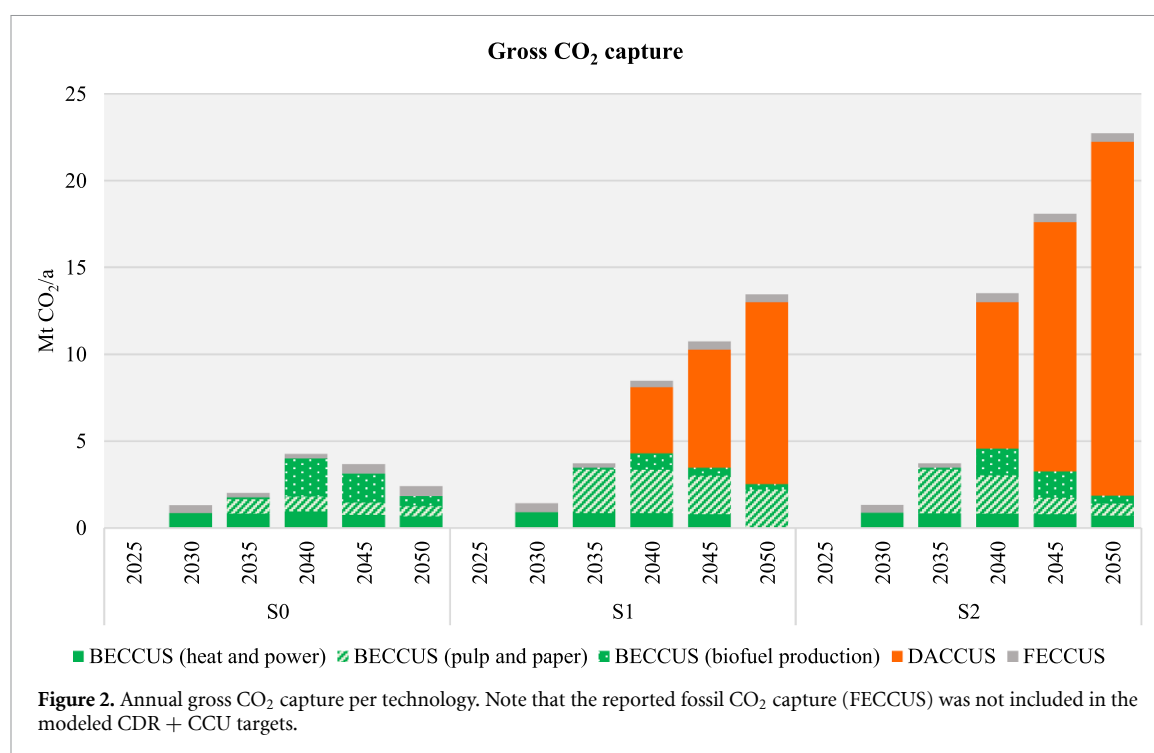
Both BECCS and DACCS are deployed in S1/S2. BECCS deployment is significantly larger in period 2035–2040 ($3.1\text{--}4.2\text{ Mt CO}_2\text{ a}^{-1}$) when compared to the S0 scenario ($1.4\text{--}3.0\text{ Mt CO}_2\text{ a}^{-1}$). However, BECCS deployment does not actually vary much between S1 and S2 despite the growing CDR targets. The greatest difference in the CDR portfolio between S1 and S2 arises from DACCS deployment, which begins in 2040. In S1, DACCS deployment increases from 3.4 to $8.3\text{ Mt CO}_2\text{ a}^{-1}$ by 2050, while in S2, 7.8 Mt CO_2 DACCS is deployed already in 2040 and deployment increases to nearly $18\text{ Mt CO}_2\text{ a}^{-1}$ by 2050. Hence, DACCS plays a much larger role in the high CDR scenario than in the low CDR scenario. Opposed to the increasing DACCS deployment trend, we notice a decline in BECCS deployment after 2040 peak levels across all scenarios. It seems that BECCS was deployed before DACCS, but BECCS was unable to grow despite the growing demand for CDR. This is explained by the modeling results, which show that the cost-efficiency of BECCS technologies seemed to depend much more on the development of other biomass-utilizing sectors than on the cost-efficiency DACCS directly. Also, the cost-optimization feature of the model targets the least-cost solution of the entire modeled system and may therefore find solutions that are not the most cost-efficient from a single actor's or sector's perspective. The deployment of BECCS vs. DACCS is further analyzed in section 3.3. On sectoral impacts.

3.2. Gross CO₂ capture

Several types of CO₂ capture technologies were modeled: biogenic CO₂ capture from heat and power production, biofuel production, and pulp and paper production; atmospheric CO₂ capture via L-DAC and S-DAC; fossil CO₂ capture from several fossil processes in industry and in energy production.

The results for technological CO₂ capture presented in figure 2 depended partially on the CDR targets and partially on the CCU targets of the scenarios. In S0, no targets for CDR were imposed, but we see some biogenic CO₂ capture from heat and power production, pulp and paper, and biofuel production. BECCUS peaks at 4 Mt CO_2 by 2040 and decreases to 1.9 Mt CO_2 by 2050. Some of it is needed for e-fuel production: $0.2\text{--}0.4\text{ Mt CO}_2$ is directed to domestic CCU in the 2030s. By 2040, $0.9\text{--}1.1\text{ Mt CO}_2\text{ a}^{-1}$ is needed for CCU. By 2050, the amount of captured CO₂ used for CCU is nearly tripled to $2.8\text{ Mt CO}_2\text{ a}^{-1}$. Interestingly, biogenic CO₂ capture was not enough in 2050 to cover the domestic need for CO₂ for e-fuels production. S0 is the only scenario where we observed CO₂ import from neighboring countries to Finland.

In S1 and S2, atmospheric CO₂ (DAC) is captured in addition to BECCUS. CO₂ capture via DAC increases from 3.8 to $10.5\text{ Mt CO}_2\text{ a}^{-1}$ by 2050 in S1. Higher CDR targets drive higher CO₂ capture via DAC in S2, in which up to 20.4 Mt CO_2 is captured by 2050. All DAC deployment is S-DAC: this is an interesting result also, as the model's choice of DAC technology affects the heat and power production sector in a very different way: S-DAC consumes multiple times more heat energy than electrical energy,



opposed to L-DAC, which consumes more electrical energy. This is observed in the sectoral impact analysis.

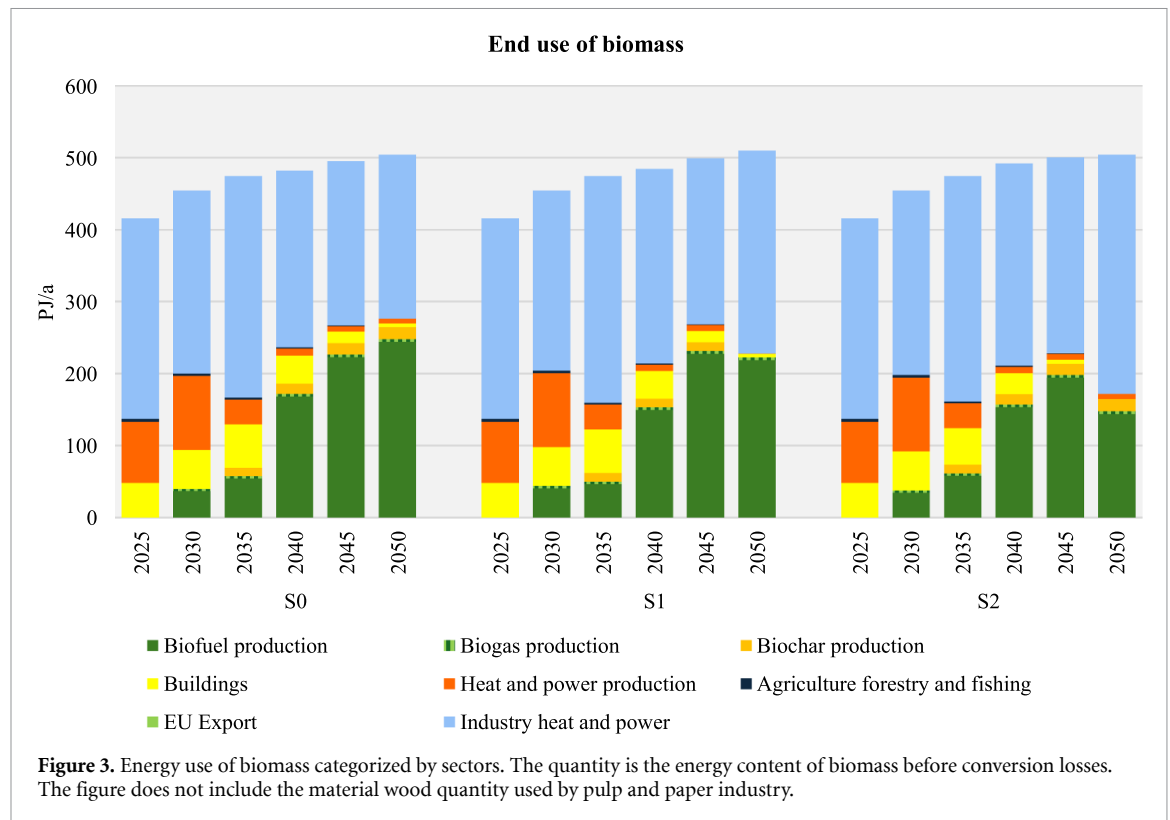
While the CDR targets affected the use of DAC by increasing the CO₂ capture capacity, biogenic CO₂ capture was rather insensitive to the CDR targets, with marginal changes in CO₂ capture capacity. However, BECCUS technologies were affected by the CDR targets in S1/S2, as we observed, e.g. much more BECCUS from pulp and paper when compared to S0. Changes in BECCUS technologies indicate that there are deeper interconnections between the modeled sectors that affect BECCUS deployment.

When looking closer at the BECCUS deployment, some interesting findings were made. BECCUS from biofuel production was deployed in every scenario, but it covered only a fraction of the total biofuel production. Considering that biofuel production volumes increased in every scenario, but BECCUS in biofuel production did not, there is a vast amount of unused BECCUS potential that was not cost-efficient for the model in the scenario solutions. Note that the model lacks the CO₂ capture option from biokerosene production. As the demand for biokerosene is driven by minimum blending requirements from ReFuelEU Aviation [34], adding this option to the model and considering it in other future studies is highly recommended. CO₂-capture from biokerosene production could affect BECCUS deployment depending on its cost-efficiency. The same observation applies to BECCUS from pulp and paper production: only a fraction of the modeled boilers in the pulp and paper industry included CO₂ capture technology.

In every scenario, we see some captured CO₂ transported to long-term geological storage, and some directed to CCU. In fact, there were no differences in CCU deployment between the scenarios, which means that the CDR + CCU targets have affected CDR deployment only, and not CCU. Therefore, CDR was more cost-efficient in our scenarios after fulfilling the minimum domestic CCU demand: ‘excess’ captured CO₂ has been transported to geological storage. Note that this result may be partially caused and/or limited by the model: trade processes for e-fuels are currently not modeled. Such markets will likely exist, which would change the allocation of CO₂ between long-term storage and CCU.

3.3. Sectoral impact assessment

Sectoral impacts were analyzed using the identified linkages of modeled CDR methods to other sectors. BECCS and biochar deployment are closely related to biomass availability and allocation between sectors; BECCS also relates to heat and power production, biofuel production, and the pulp and paper industry. DACCS is largely affected by available renewable electricity and heat, and therefore closely related to both production and end use of domestic heat and power. Lastly, captured CO₂ is linked to storage and utilization of CO₂, such as e-fuel production and consumption.



3.3.1. Biomass consumption

The energy use of biomass increases across all scenarios from 416 PJ a⁻¹ in 2025 up to 510 PJ a⁻¹ in 2050 by an average of 22% (figure 3). While increasing biomass demand is seen in all scenarios, the use of biomass differs.

In our results, biomass is used in industry, heat and power production, buildings, agriculture, biofuel and biogas production, biochar production, and some are exported to the EU. Most of the biomass is used in industry for process heat and electricity in all scenarios throughout the time series; however, the industrial use varies between 220–330 PJ a⁻¹, and no clear trend is seen in biomass use in industry.

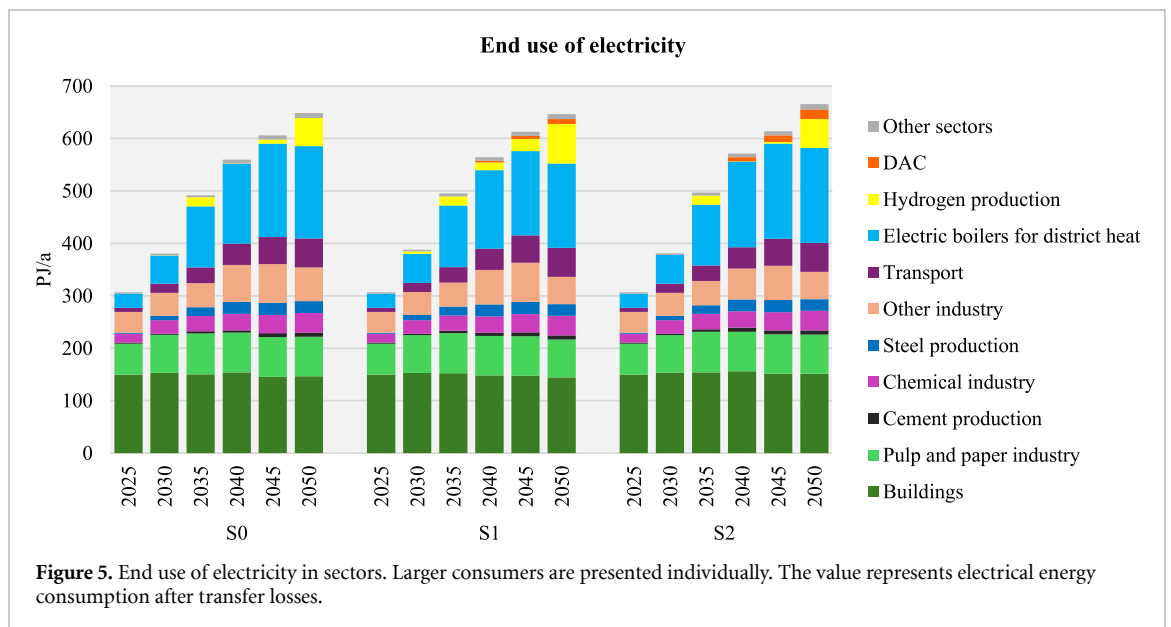
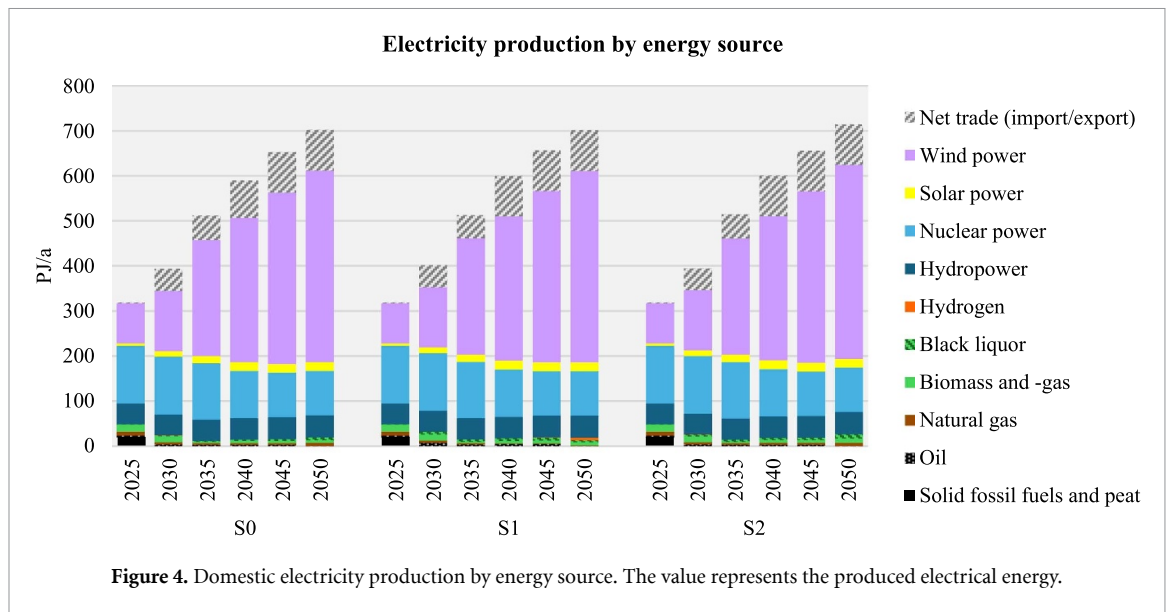
A significant amount of solid biomass is used directly in buildings for heating. This declines over time, approaching zero by 2050. Use of biomass in the heating of buildings is replaced with direct electricity and heat pumps in all scenarios. Also, the agriculture, forestry, and fishing sector consumes some biomass for energy, but their consumption decreases to zero by 2050 in all scenarios.

District heat production and electricity generation are also major biomass users in the 2020s, but their biomass use drops significantly already in 2035 and continues to decline, approaching zero by 2050. As the biomass use in heat and power production starts to decline in 2035, biochar production increases. In 2035, 12 PJ of biomass is consumed annually for biochar production. It remains constant in the S1 scenario, but in S0 and S2, it increases up to 17 PJ a⁻¹ by 2050.

The clearest differences in biomass use are seen in biofuel production. In the cost-optimal scenario S0, biomass use in biofuel production increases from 37 PJ a⁻¹ in 2030–244 PJ a⁻¹ by 2050, whereas in S1/S2 the biomass use is somewhat less, although the increasing consumption trend occurs in every scenario. An increasing trend is also visible in the domestic end use and export of liquid biofuels. Of the two, domestic end use does not vary in the scenarios and is therefore insensitive to the CDR targets set in S1/S2. It is the export of biofuels that contributes to the differences in biomass use in biofuel production seen in figure 3. Export of biofuels is the largest in S0, and smallest in S2. Therefore, with higher CDR targets, we see less domestic biofuel production and export in our results. Overall, direct combustion of biomass for energy in industry, buildings, and the energy sector decreases while conversion to biofuels increases in the long term.

3.3.2. Electricity production

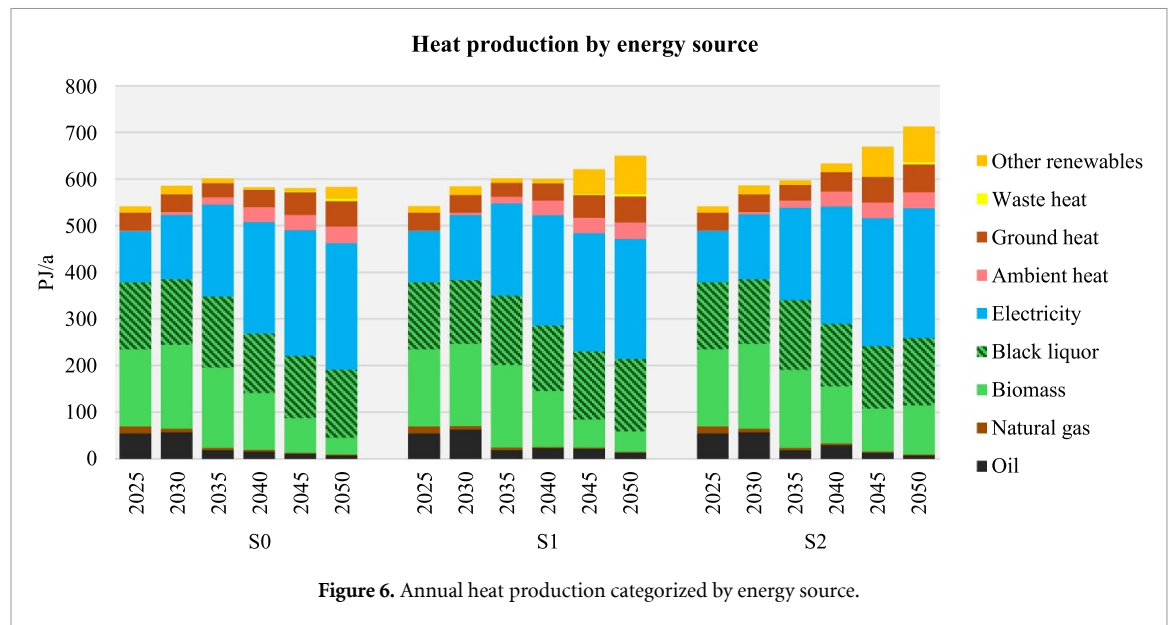
Annual electricity production more than doubled from 319 PJ (89 TWh) in 2025 up to 714 PJ (198 TWh) by 2050 (figure 4). General trend in electricity production is the same in each scenario: energy sources used for domestic electricity production do not vary much between the scenarios. The



overall increase in electricity production in the scenarios is generated mainly by vast increases in wind power. In addition to wind, electricity production in the long term relies heavily on other renewable or non-fossil sources like nuclear, hydropower, and solar power, together with small quantities of biomass and biogas. The remaining fossil fuel-based production is in a marginal role already in 2030. However, cross-border net trade of electricity is positive across the time series and all the scenarios, which means that domestic production is not enough, or cost-efficient enough, to cover domestic electricity demand. The only notable difference between the scenarios is the $>10 \text{ PJ a}^{-1}$ (3 TWh a^{-1}) higher electricity production volume in 2050 in S2 when compared to S0/S1, despite the large differences in DAC deployment.

The reasons for the large increase in electricity production volumes can be detected from the end use of electricity (figure 5). The growing electricity demand is driven by electrification of industrial processes, transport, and district heat production, along with hydrogen production. Annual electricity demand of DAC in 2050 is 9 PJ (2.6 TWh) in S1 and 18 PJ (5.1 TWh) in S2, which are only fractions of the total demand. For example, electrification of transport requires 55 PJ a^{-1} (15 TWh) in 2050, which is three times more than the electricity demand of DAC.

The low electricity demand of DAC is explained by the technology choice made by the model. All of the DAC is S-DAC, which consumes energy mainly in the form of heat. If L-DAC were deployed, we would see much larger electricity demand by DAC than with S-DAC.



3.3.3. Heat production

Total heat production, including process heat of industry, is presented in figure 6. Heat production in S0 stabilizes to a level of 580 PJ a^{-1} (155 TWh a^{-1}). However, in S1 and S2, heat production increases in the long term by 23%. In S1, total production in 2050 is 640 PJ a^{-1} (178 TWh a^{-1}), and in S2, 703 PJ a^{-1} (195 TWh a^{-1}).

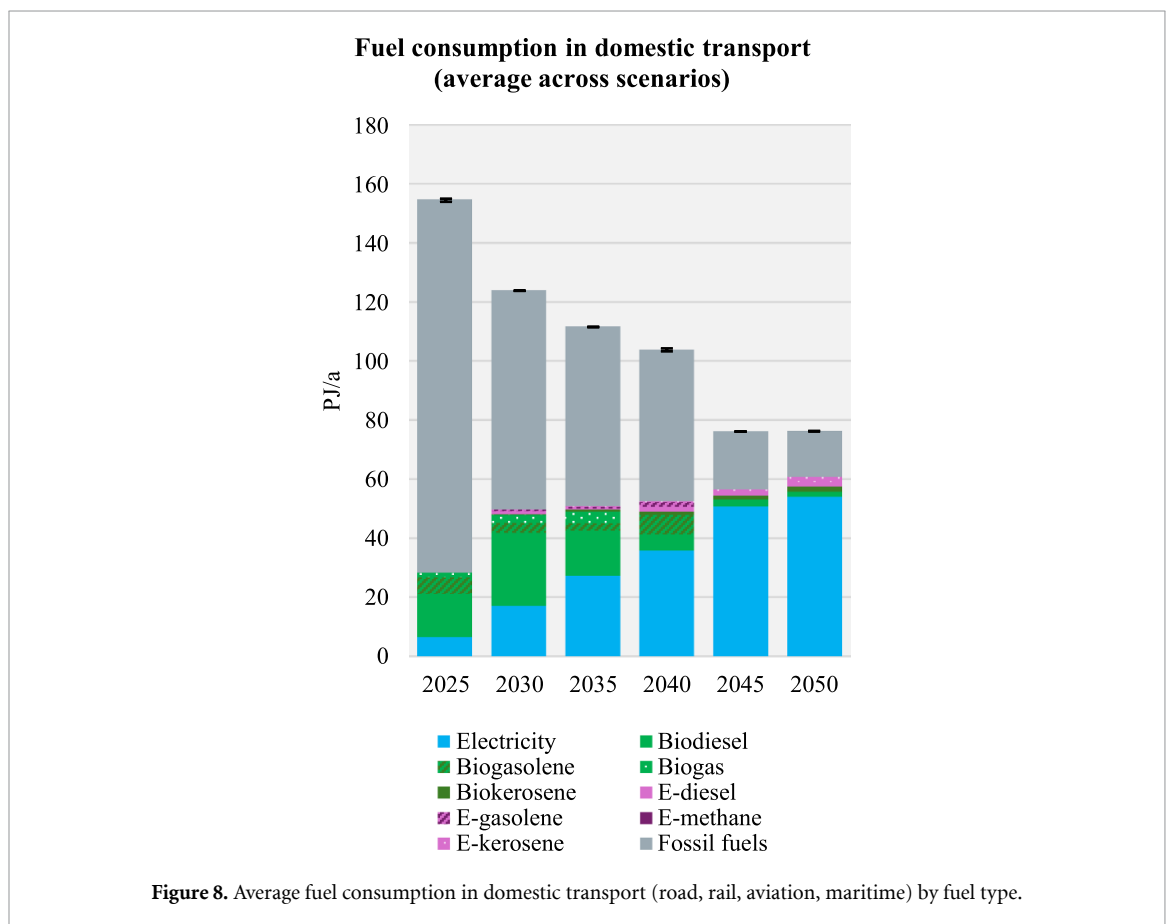
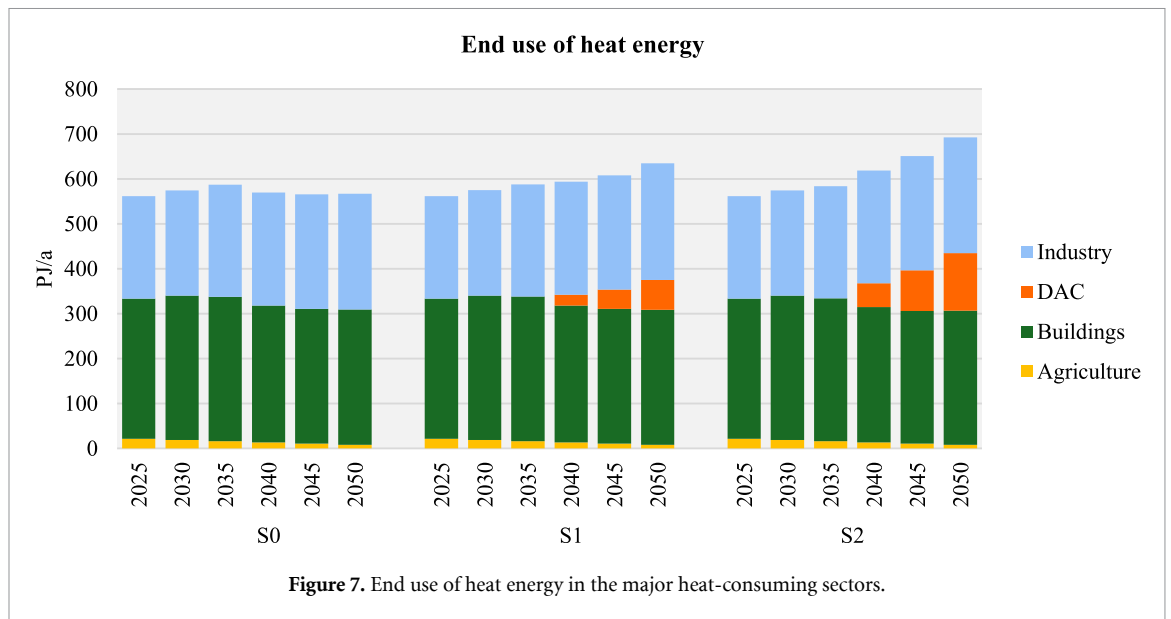
Electrification of heat production is a major trend across the scenarios. Heat production by electric boilers increases 134%–153% in the scenarios, the least in S1 and the most in S2. Also, the use of air and ground heat pumps in heat production increases throughout the scenarios in the long term. At the same time, the use of biogenic fuels in heat production decreases 21%–43%, the least in S2 and the most in S0. Industry used more biomass for process heat and electricity in S2 than in S0/S1, and this is seen in the heat production: although use of biogenic fuels decreases over time, S2 still uses the largest amount of solid biomass for industry process heat in the long term. A significant increase in the use of other renewables, most of it being geothermal, is seen in S1/S2 when compared to S0.

End use of heat increased only due to the S-DAC process (figure 7). Heat demand by agriculture, buildings, and industry does not change in the scenarios. Hence, the increase in total heat production is caused by DACCS and is directly proportional to DACCS deployment. The increased heat demand in S2 was covered by renewable energy sources, most of it being direct electricity, geothermal, and biomass. A closer look at the individual technologies revealed that the heat demand of S-DAC was almost solely covered by electric boilers, while biomass boiler usage increased in industry. Note that the model was missing some waste heat sources and heat pumps, which may have been cost-efficient solutions to provide fossil-free heat.

3.3.4. Transport

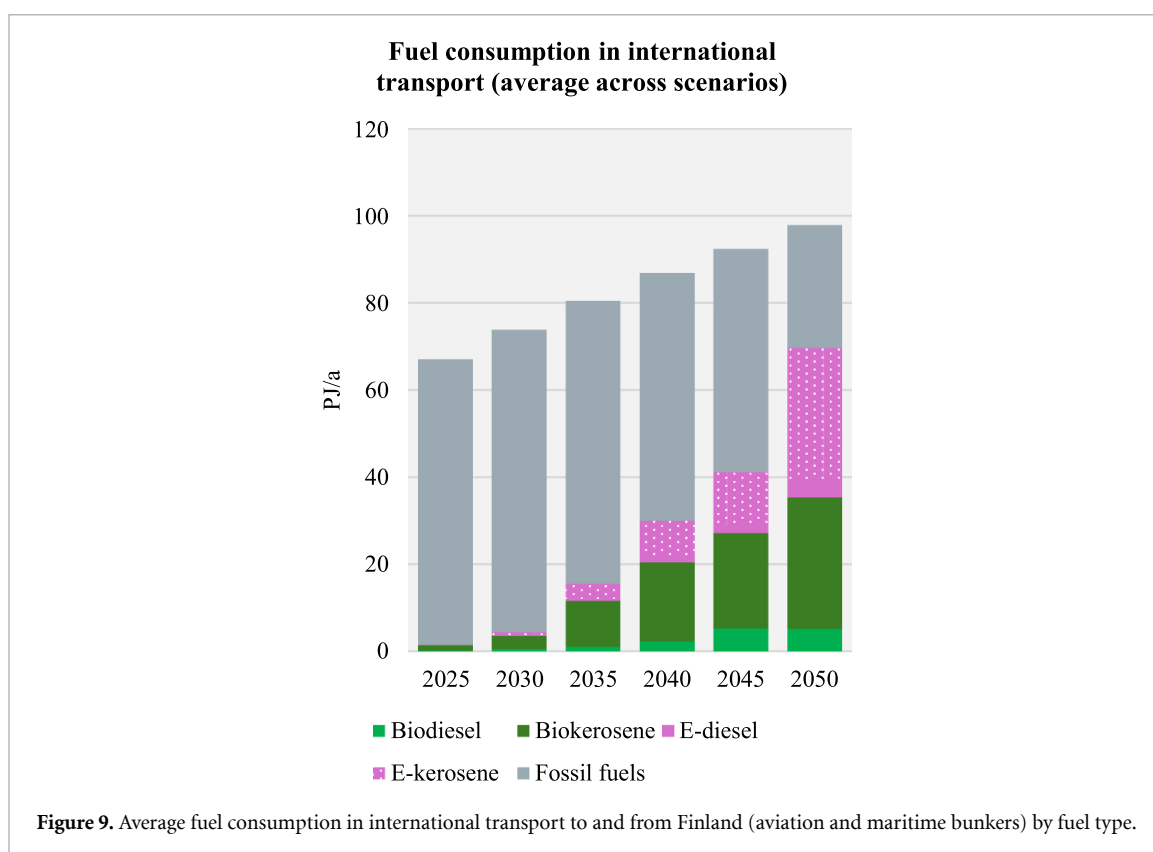
Road transport sector undergoes near-complete electrification by 2050. Fossil fuels are replaced by electricity, biofuels, and e-fuels. Electricity becomes the primary energy source for passenger transport, while heavy-duty transport still relies partially on diesel in 2050. Direct hydrogen utilization does not occur in any of the scenarios; however, hydrogen enters the transport sector indirectly through the production and use of e-fuels. Overall energy consumption in domestic transport decreases from 155 PJ a^{-1} to 76 PJ a^{-1} by 2050 due to the higher energy efficiency of electric vehicles (figure 8). There are no significant differences in transport demand or in transport energy consumption between the scenarios.

A closer look at the fuel component level consumption reveals a temporary increase in biofuel demand in domestic transport. This is due to the renewable fuel obligation that aims at increasing the minimum share of renewable liquid and gaseous biofuels, electricity, and e-fuels in the total energy distributed to public fueling or charging infrastructure in Finland [53]. The minimum renewable energy requirement increases to 34% by 2029, after which it stays at a constant level. As the car fleet electrifies, an ever higher amount of renewable electricity is accounted towards fulfilling the renewable fuel obligation. Also, electrification drives the overall decreasing consumption of liquid and gaseous fuels. As the minimum obligation level remains constant after 2030, increasing amount of renewable electricity



replaces biodiesel, biogas, and bioethanol, of which demand decreases after 2035. Hence, biofuel demand in domestic car transport becomes marginal by 2050. However, we see an increase in biokerosene and e-kerosene demand for both domestic and international aviation by 2050, which is induced by EU-level regulation [34].

Energy consumption in international transport to and from Finland increases over time in all scenarios due to assumptions about the demand for aviation (figure 9). International transport is still fully powered by fossils in 2020. In the long term, most of the fossil fuels is replaced by biofuels and e-fuels in all the scenarios. The transition to low and zero-carbon fuels is driven by CCU targets in our scenarios, which were based on the requirements of EU-level regulation [34, 35]. Despite the flexibility given



to the model to blend more than the minimum required amount of e-fuels and biofuels into the fuel blends, there are no differences in consumption patterns between the scenarios. Thus, neither high or low CDR targets have affected the transport sector in any way. Geological storage of CO₂ has been a more cost-efficient solution for the use of captured CO₂ than producing more e-fuels for transport.

3.4. CO₂ emission development

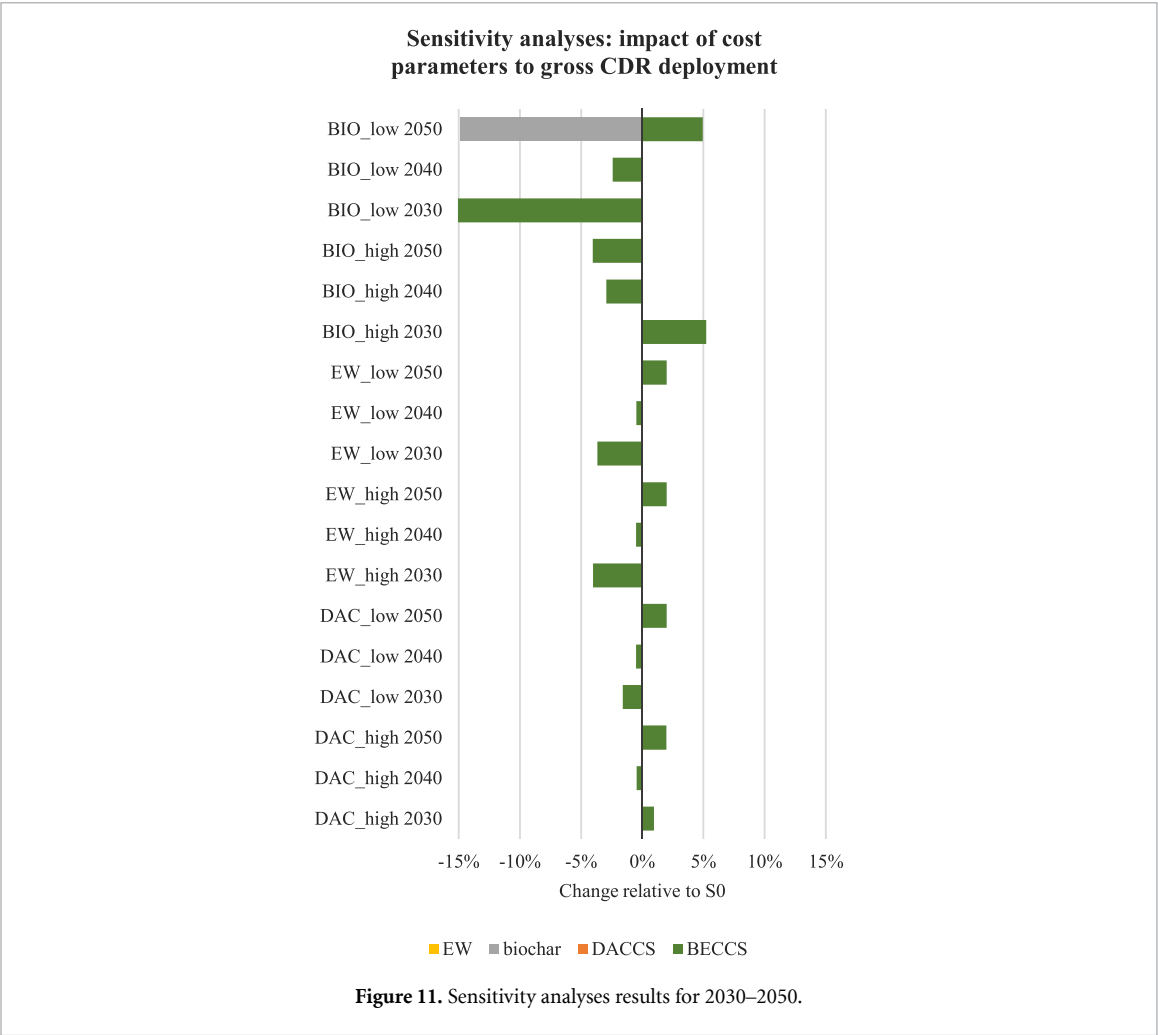
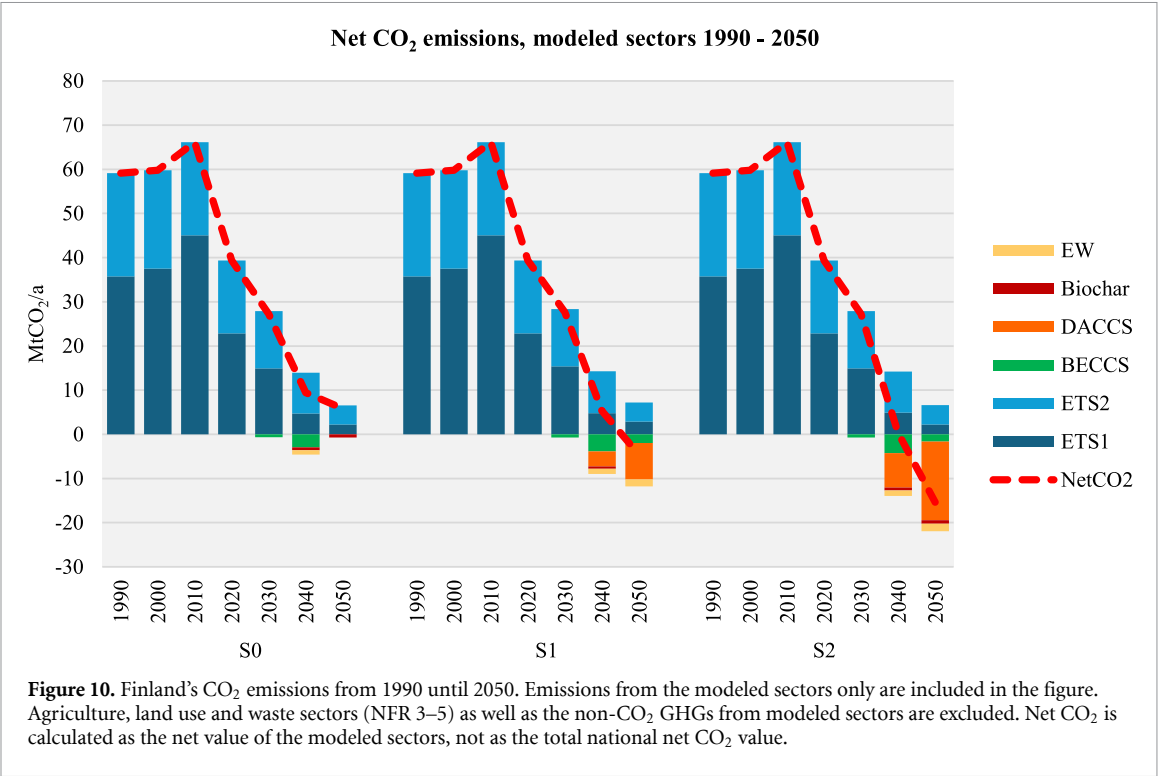
The overall CO₂ emission development is very different in each scenario in the long term (figure 10). In S0, modeled sectors did not reach net zero emissions in Finland by 2050, although on the EU-level, the model achieved the required modeling target. However, nearly 5 Mt CO₂ of negative emissions were cost-efficiently deployed in 2040. In S1, the national net zero CO₂ emissions in the modeled sectors was reached in 2045, while in the high CDR scenario S2, it was reached even earlier in 2040.

CO₂ emissions from ETS1 decreased more rapidly than from ETS2 despite the fact that the carbon price was the same for both ETS systems from 2035 onwards: faster decarbonization could be achieved with higher CO₂ prices or more stringent emission allowance limits. Emissions from ETS1 and ETS2 followed the same path in every scenario until 2035. Interestingly, both ETS1 and ETS2 emissions are somewhat higher in S1/S2 than in S0 after 2040, when CDR deployment increased. Despite that, national net CO₂ emissions from modeled sectors decreased much more rapidly with any CDR target than without. A higher CDR target seems beneficial for Finland's emission development: energy-related CO₂ emissions could be fully compensated with a versatile CDR portfolio together with rapid CO₂ reductions in ETS sectors.

As the LULUCF, agriculture, and waste sectors were not included in the modeling, the net zero in our scenarios only applies to the energy sectors modeled, and not to the total GHG emission development. As the biomass supply in the scenarios is assumed to stay at a high level, around 80 Mm³ total harvest [8, 15], the LULUCF sector will likely remain an emission source of around 7 Mt CO_{2eq} until 2050. This would mean that even with significant deployment of BECCS and DACCS, Finland would reach climate neutrality by 2050 only in the high CDR scenario S2.

3.5. Sensitivity analyses

In the sensitivity analyses, DAC investment costs, EW carbon sequestration cost, and biomass feedstock costs, were varied $\pm 25\%$. We observed the relative change in gross CDR deployment of each CDR method relative to the result in S0. Across the sensitivity analyses, we detected only changes in BECCS and biochar deployment; no other CDR deployment was affected (figure 11). Lower biomass feedstock



cost in BIO-low resulted initially in lower BECCS deployment. In long the term, BECCS deployment increased as biomass feedstock was freed due to lower biochar deployment level. It seems that lower biomass feedstock cost increases biomass use in other applications than BECCS and biochar, but in the long term, BECCS benefits from lower feedstock costs. Higher biomass feedstock cost in Bio-high showed a decreasing trend in BECCS deployment, while biochar deployment was not changed by higher feedstock costs. In terms of gross capture of CO₂, the changes are small: when expressed in CO₂ units, the change in BECCS deployment translates to a maximum change of 130 kt CO₂. The changes in BECCS that occurred in scenario runs, where DAC or EW cost were varied, are explained by the system's interconnectivities: biomass, biofuel, and electricity trade together with EU-ETS systems link the modeled regions' energy systems together. Changes occurring elsewhere in the modeled system, i.e. the rest of Europe, may have an indirect impact in Finland. The minor impact of the variation of DAC investment costs on the results may be explained by the high significance of the energy cost in the DACCS total cost. Here, the cost of electricity and heat is endogenously defined by the model. On the other hand, the initial assumption on the EW cost in this study is rather low, so even increasing it by 25% does not threaten EW's cost-efficiency in comparison to many other mitigation solutions.

4. Conclusions and discussion

The goal of this work was to model scenarios for CDR deployment in Finland and to analyze the impacts on the energy system. Four different CDR technologies were considered: BECCS, DACCS, biochar soil amendment, and EW of mining waste rocks. The Finnish case study enables drawing conclusions applicable also to other regions. However, some of the conclusions can be specific to countries with similar energy systems, characterized by integration of electricity production and heating sectors and by large-scale biomass use.

We modeled three scenarios: the reference scenario S0 without CDR targets, and two scenarios, S1 and S2, with low or high CDR targets, respectively. The S0 reflects the current status of the Finnish and European climate policy, where no separate targets have been set for permanent, technological CDR in reaching the net-zero emissions. In S1 and S2, CDR targets were set based on the results from previous long-term scenarios for Finland in which BECCS and/or DACCS were utilized [8, 30]. Minimum targets for CCU were set as an e-fuel mandate based on the current EU regulations [33–35].

Several different BECCS and DACCS technologies were included in the modeling (see SI), and their potentials were limited by the availability of biomass resources and the energy constraints. The biochar and EW potentials were limited due to uncertainties in their potentials and costs, as elaborated in section 2.3. Because the EW potential allowed was fully utilized already in the reference scenario S0, the CDR targets affected only the deployment of biochar, BECCS, and DACCS.

Based on our results, the national net-zero target will be easier to reach with CDR deployment than without: without a separate CDR target, the net-zero target was not achieved in the modeled energy-consuming and producing sectors by 2050. With the higher CDR target of 25 Mt CO₂ a⁻¹ by 2050 (S2) and the lower target of 15 Mt CO₂ a⁻¹ by 2050, the modeled net-zero CO₂ was reached by 2040 and 2045, respectively. However, when considering the GHG emissions from the non-modeled sectors, the perspective changes: if the LULUCF sector remains as an emission source of around 7 Mt CO_{2eq} as estimated in [8], Finland would need even a more than 25 Mt CO₂ a⁻¹ CDR deployment to become climate neutral by 2050. Thus, the introduction of BECCS and biochar should be done without increasing wood harvests, as the impacts of wood harvests on the GHG emissions are seen in the LULUCF sector.

The above conclusions are also relevant for many other regions; a separate target for permanent CDR would increase the probability of reaching net-zero GHG emissions by 2050, but the LULUCF sink needs to be strengthened simultaneously. The need for both permanent CDR and natural sinks is also reflected in the European Commission scenarios for reaching 2040 climate targets in the EU, as well as in the recommendations of the European Scientific Advisory Board for Climate Change [54, 55].

Interestingly, in this study, BECCS and biochar were rather insensitive to the CDR targets, and most of the growth in CDR deployment originated from DACCS. The results also showed complex connections among BECCS, biochar, and DACCS when modeled as a part of the Finnish energy system. Biomass feedstock availability and allocation were at the core of these interconnections.

For the first time in the Finnish scenario modeling studies, both S-DAC and L-DAC technologies were modeled. The results indicate that S-DAC could be a more cost-efficient choice for Finland. S-DAC is characterized by higher heat demand and lower electricity demand than L-DAC. This has important implications for the Finnish energy system, which is characterized by significant heat consumption and centralized heat production via district heating systems. Typically, one would expect significant impact

on the electricity demand when deploying L-DAC, but in our results, the electricity demand was only marginally increased despite the high DAC deployment. Instead, heat production and demand underwent significant changes with S-DAC deployment.

The use of S-DAC increased the total domestic heat demand. Although the heat for S-DAC was mostly generated with electric boilers, the capacity of electric boilers did not increase similarly with S-DAC deployment. Instead, more biomass boilers were added in industry, to replace the heat from electric boilers directed for S-DAC. Thus, biomass acted as a buffer to provide additional fossil-free heat. This is an interesting finding, which may also reflect cost- or energy efficiency aspects of electric boilers, or bottlenecks in electricity generation. It is worth noting that our model did not include potential new investments in data centers, nor many currently existing waste heat sources, which may have affected the results, and should be further studied. Similar energy system impacts are likely encountered in other regions, but specific studies for each energy system with individual characteristics are needed, as the heating sectors differ considerably in different regions.

A general trend across sectors, transport, heating and industry, was electrification, which occurred throughout the scenarios. Electrification is clearly a prerequisite for the decarbonization of Finland, as also shown in the recent national baseline scenario [8]. Electrification increased electricity demand substantially over the modeled time horizon, and a significant share of electricity was used in electric boilers to produce fossil-free heat. The electricity generation profile was the same in every scenario, with a large share of production based on fossil-free production of wind, nuclear, hydro, and solar. Every scenario depended on electricity imports, reflecting the challenges of scaling up fossil-free domestic electricity generation in a cost-efficient way. Neighboring regions were able to supply electricity at lower costs, particularly during the winter months, when demand peaks due to need for heating.

Another clear trend was the increasing use of biomass due to rising biofuel production and higher demand in the industry. By contrast, biomass use decreased in electricity generation, and also somewhat in heat production (excluding industrial high-temperature heat). The supply of biomass was limited in terms of domestic wood harvest and imports from the Baltic Sea region. Domestic wood harvesting was based on recent estimates [15] of the pulp and paper industry demand, which is a major wood consumer in Finland.

Biofuel production is expected to increase in Europe, as the transport sector, especially aviation and navigation, pursues decarbonization through biofuel and e-fuel use in addition to electrification due to EU regulations [33–35]. Import of biofuels is an option, but in our scenarios, domestic production in Finland was a cheaper alternative to satisfy biofuel demand. In fact, large amounts of biofuels were exported. The increasing e-fuel demand drives technological CO₂ capture in our results. Domestic biogenic CO₂ capture of less than 5 Mt CO₂ a⁻¹ could be enough for the domestic future e-fuel demands of aviation and navigation until 2050.

In recent studies, BECCS in the pulp and paper industry has been the dominant source of CO₂ in Finland, while DACCS has played a small role [8, 30]. In this study, despite the increasing use of biomass, investments in BECCS and biochar did not increase significantly. On the other hand, earlier regional studies did not identify S-DAC technology at all [8–11, 15, 30], whereas in this study it had a dominant role in the scenarios, indicating that the model found S-DAC to be a cost-efficient solution. The differing results can also be partly explained by differences in modeling tools used, scenario definitions, costs, and input/output assumptions of modeled technologies. Inefficiencies may also play a role: introducing CO₂ capture to bioenergy processes has been shown to decrease process efficiency and to reduce the amount of sellable products such as electricity [8, 12]. It is important to note that results from scenario studies should not be interpreted as future predictions, but they can provide information on the scale of solutions needed.

Cost and energy efficiency assumptions affect the results, and studying those variables more in-depth for BECCS and biochar processes would be of interest in future studies, also bringing up the CO₂ capture potential in biofuel production. If domestic wood harvesting decreased in the future, e.g. due to decreased demand for wood in forest industries or increased efforts to restore the carbon sinks of forests, this could affect the biomass use. Lower supply could increase biomass prices, which could make BECCS and biochar more expensive due to increasing feedstock costs.

Our study showed that scaling up energy-intensive DACCS and biomass-dependent BECCS will have broad impacts across the energy system. Trade-offs come with every solution, and the impacts of CDR deployment on future energy systems need to be further studied in different regions. Scaling up CDR deployment in Finland and elsewhere would need a significant amount of investment, and cost-efficient CDR solutions should be prioritized. In addition, significant investments in the surrounding energy system, CO₂ logistics and geological storage capacity are needed. The investors would need a clear visibility that permanent CDR will play a role in the decarbonization. Thus, binding targets for permanent

CDR, investment supports, and setting a price for the removed CO₂, e.g. through careful introduction to EU-ETS [56] or other instruments are needed. It is also evident that many climate measures in the land use sector are more cost-efficient than deploying technological CDR solutions, and should be executed in parallel [54]. Decarbonization of energy systems with limited resources will raise questions about cost-efficient allocation of those limited resources. It is important to study and identify possible pitfalls, where technological CDR deployment may fail due to systemic inefficiencies and lack of fossil-free energy sources, and what is the role of restoring natural carbon sinks and deploying more cost-efficient nature-based CDR on the path towards net zero.

Acknowledgment

Many thanks to our colleague Antti Lehtilä for his insights about the model. The work was funded by the financial support for the green transition by the European Union (No. 151, P5C1I2, Next Generation EU, project REPower-CEST).

Data availability statement

The data cannot be made publicly available upon publication because they are not available in a format that is sufficiently accessible or reusable by other researchers. The data that support the findings of this study are available upon reasonable request from the authors.

Supplementary data 1 available at <https://doi.org/10.1088/2753-3751/ae57b0/data1>.

Supplementary data 2 available at <https://doi.org/10.1088/2753-3751/ae57b0/data2>.

Funding

The work was funded by the financial support for the green transition by the European Union (No. 151, P5C1I2, Next Generation EU, project REPower-CEST).

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